

Exercises Chapter 1: The Time Value of Money

Additional Exercises

The Timeline

The Three Rules of Time Travel

1) Consider the following alternatives:

- i. \$150 received in one year
 - ii. \$220 received in five years
 - iii. \$330 received in ten years
- a. Rank the alternatives from most valuable to least valuable if the interest rate is 11% per year.
 - b. What is your ranking if the interest rate is only 4% per year?
 - c. What is your ranking if the interest rate is 15% per year? (BDM 4.6)

a. $PV = \frac{\$150}{1.11^1} = \135.14

$$PV = \frac{\$220}{1.11^5} = \$130.56$$

$$PV = \frac{\$330}{1.11^{10}} = \$116.22$$

Option i. > option ii. > option iii.

b. $PV = \frac{\$150}{1.04^1} = \144.23

$$PV = \frac{\$220}{1.04^5} = \$180.82$$

$$PV = \frac{\$330}{1.04^{10}} = \$222.94$$

Option iii. > option ii. > option i.

c. $PV = \frac{\$150}{1.15^1} = \130.43

$$PV = \frac{\$220}{1.15^5} = \$109.38$$

$$PV = \frac{\$330}{1.15^{10}} = \$81.57$$

Option i. > option ii. > option iii.

2) Suppose you invest \$1000 in an account paying 6% interest per year. (BDM 4.7)

- a. What is the balance in the account after 3 years? How much of this balance corresponds to “interest on interest”?
- b. What is the balance in the account after 35 years? How much of this balance corresponds to interest on interest?

a. $\text{Balance after 3 years} = \$1,000 \times 1.06^3 = \$1,191.02$

$$\text{Interest} = \$1,000 \times 0.06 \times 3 = \$180$$

$$\text{Interest on interest} = \$1,191.02 - \$1,000 - \$180 = \$11.02$$

- b. *Balance after 35 years* = $\$1,000 \times 1.06^{35} = \$7,686.09$
Interest = $\$1,000 \times 0.06 \times 35 = \$2,100$
Interest on interest = $\$7,686.09 - \$1,000 - \$2,100 = \$4,586.09$

Valuing a Stream of Cash Flows

- 3) Suppose you receive \$100 at the end of each year for the next three years. (BDM 4.11)

- If the interest rate is 7%, what is the present value of these cash flows?
- What is the future value in three years of the present value you computed in (a)?
- Suppose you deposit the cash flows in a bank account that pays 7% interest per year. What is the balance in the account at the end of each of the next three years (after your deposit is made)? How does the final bank balance compare with your answer in (b)?

a. $PV = \frac{\$100}{1.07^1} + \frac{\$100}{1.07^2} + \frac{\$100}{1.07^3} = \$93.46 + \$87.34 + \$81.63 = \$262.43$

b. $FV = \$262.43 \times 1.07^3 = \321.49

- c. Year 1: \$100

Year 2: $100 \times 1.07 + 100 = \207

Year 3: $100 \times 1.07^2 + 100 \times 1.07 + 100 = \321.49

Calculating the Net Present Value

- 4) You have been offered a unique investment opportunity. If you invest \$15,000 today, you will receive \$750 one year from now, \$2250 two years from now, and \$15,000 ten years from now. (BDM 4.14)

- What is the NPV of the opportunity if the interest rate is 12% per year? Should you take the opportunity?
- What is the NPV of the opportunity if the interest rate is 8% per year? Should you take it now?

a. $NPV = -\$7,706.87$. Since the $NPV < 0$, don't take it.

b. $NPV = -\$5,428.65$. Since the $NPV < 0$, don't take it.

Perpetuities and Annuities

- 5) The British government has a consol bond outstanding paying £300 per year forever. Assume the current interest rate is 4% per year. (BDM 4.18)

- a. What is the value of the bond immediately after a payment is made?
 - b. What is the value of the bond immediately before a payment is made?
- a. The value of the bond is equal to the present value of the cash flows. By the perpetuity formula:

$$\frac{£300}{0,04} = £7,500$$
 - b. The value of the bond is equal to the present value of the cash flows. The cash flows are the perpetuity plus the payment that will be received immediately.

$$PV = 300/0.04 + 300 = £7,800$$
- 6) Your brother has offered to give you \$120, starting next year, and after that growing at 2.8% for the next 20 years. You would like to calculate the value of this offer by calculating how much money you would need to deposit in the local bank so that the account will generate the same cash flows as he is offering you. Your local bank will guarantee a 6.3% annual interest rate so long as you have money in the account. (BDM 4.31)
- a. How much money will you need to deposit into the account today?
 - b. Deleted.
- a. $120 * (1 / (0.063 - 0.028)) * (1 - (1.028 / 1.063)^{20}) = 1,673.44$

Solving for the Cash Payments

- 7) You are 27 years old, and decide to save \$15,000 each year (with the first deposit one year from now), in an account paying 9% interest per year. You will make your last deposit 38 years from now when you retire at age 65. During retirement, you plan to withdraw funds from the account at the end of each year (so your first withdrawal is at age 66). What constant amount will you be able to withdraw each year if you want the funds to last until you are 85? (BDM 4.44)

Step 1: Calculate the *PV* of the annuity (at age 27):

$$PV = \$15,000 \times \frac{1}{0.09} \left(1 - \frac{1}{1.09^{38}} \right) = \$160,362.2947$$

In *FV* at the age of 65, this is equal to $\$160,362.2947 \times 1.09^{38} = \$4,239,446.743$

Step 2: Calculate the value of the payment that we can cash out for an annuity that will pay 20 times, i.e. from the day we turn 66 to the day we turn 85:

$$C = \frac{\$4,239,446.743}{\frac{1}{0.09} \left(1 - \frac{1}{1.09^{20}} \right)} = \frac{\$4,239,446.743}{9.128545669} = \$464,416.45$$

The Internal Rate of Return

- 8) Suppose you invest \$2000 today and receive \$9750 in five years. (BDM 4.48)
- What is the IRR of this opportunity?
 - Suppose another investment opportunity also requires \$2000 upfront, but pays an equal amount at the end of each year for the next five years. If this investment has the same IRR as the first one, what is the amount you will receive each year?

a. IRR is that r that solves:

$$\$2,000 = \frac{\$9,750}{(1+r)^5} \rightarrow \left(\frac{\$9,750}{\$2,000} \right)^{\frac{1}{5}} - 1 = 0.3728 = 37.28\%$$

b. Timeline:

$$\$2,000 = \frac{X}{IRR} \left(1 - \frac{1}{(1+IRR)^5} \right)$$

Thus,

$$X = \frac{\$2,000}{\frac{1}{0.3728} \left(1 - \frac{1}{1.3728^5} \right)} = \$938.01$$

- 9) A local bank is running the following advertisement in the newspaper: "For just \$3000 we will pay you \$180 forever!" The fine print in the ad says that for a \$3000 deposit, the bank will pay \$180 every year in perpetuity, starting one year after the deposit is made. What interest rate is the bank advertising (what is the IRR of this investment)? (BDM 4.50)

The payments are a perpetuity, so:

$$PV = \frac{\$180}{r}$$

$$\$3,000 = \frac{\$180}{r}$$

$$r = \frac{180}{3,000} = 0.06 = 6\%$$

- 10) You are considering investing in a security that will pay you \$80 in interest-only at the end of each of the next 10 years. If this security is currently selling for \$588.81, then the IRR for investing in this security is closest to: (BDM Test bank 4)

A) 6.0%.

B) 7.0%.

C) 6.5%.

D) 5.0%.

Exercises Chapter 2: Investment Decision Rules**Additional Exercises****NPV and Stand-Alone Projects**

- 1) You are considering investing in a start-up company. The founder asked you for \$300,000 today and you expect to get \$1,030,000 in 11 years. Given the riskiness of the investment opportunity, your cost of capital is 30%. What is the NPV of the investment opportunity? Should you undertake the investment opportunity? Calculate the IRR and use it to determine the maximum deviation allowable in the cost of capital estimate to leave the decision unchanged. (BDM 7.2)

$$NPV = -\$300,000 + \frac{\$1,030,000}{1.30^{11}} = -\$242,527.47$$

You should not undertake the investment opportunity.

$$IRR = \left(\frac{\$1,030,000}{\$300,000} \right)^{\frac{1}{11}} - 1 = 0.1187 = 11.87\%$$

A drop in the cost of capital of up to $30\% - 11.87 = 18.13\%$ would not change the decision.

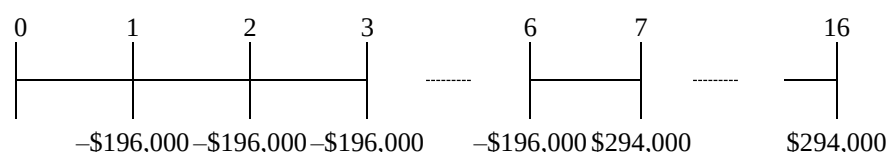
- 2) You are considering opening a new plant. The plant will cost \$100 million upfront. After that, it is expected to produce profits of \$30 million at the end of every year. The cash flows are expected to last forever. Calculate the NPV of this investment opportunity if your cost of capital is 8%. Should you make the investment? Calculate the IRR and use it to determine the maximum deviation allowable in the cost of capital estimate to leave the decision unchanged. (BDM 7.3)

NPV = $-100 + 30/8\% = \$275$ million. Yes, make the investment.

IRR: $0 = -100 + 30/IRR$. IRR = $30/100 = 30\%$. Okay as long as cost of capital does not go above 30%.

- 3) FastTrack Bikes, Inc. is thinking of developing a new composite road bike. Development will take six years and the cost is \$196,000 per year. Once in production, the bike is expected to make \$294,000 per year for 10 years. Assume the cost of capital is 10%. (BDM 7.6)
- Calculate the NPV of this investment opportunity, assuming all cash flows occur at the end of each year. Should the company make the investment?
 - By how much must the cost of capital estimate deviate to change the decision?
 - What is the NPV of the investment if the cost of capital is 15%?

a. Timeline:



$$NPV = \left[-\$196,000 \times \frac{1}{0.10} \left(1 - \frac{1}{1.10^6} \right) \right] + \frac{\left[\$294,000 \times \frac{1}{0.10} \left(1 - \frac{1}{1.10^{10}} \right) \right]}{1.10^6}$$

$$NPV = -\$853,631.10 + \$1,019,723.70 = \$166,092.60$$

b. Setting the $NPV = 0$ and solving for r (using a spreadsheet) the answer is $IRR = 12.66\%$. A drop in the cost of capital of more than $12.66\% - 10\% = 2.66\%$ would change the decision.

$$c. \quad NPV = \left[-\$196,000 \times \frac{1}{0.15} \left(1 - \frac{1}{1.15^6} \right) \right] + \frac{\left[\$294,000 \times \frac{1}{0.15} \left(1 - \frac{1}{1.15^{10}} \right) \right]}{1.15^6}$$

$$NPV = -\$741,758.61 + \$637,907.14 = -\$103,851.47$$

The Internal Rate of Return Rule

- 4) You have been offered a very long term investment opportunity to increase your money one hundredfold. You can invest \$500 today and expect to receive \$50,000 in 40 years. Your cost of capital for this (very risky) opportunity is 16%. What does the IRR rule say about whether the investment should be undertaken? What about the NPV rule? Do they agree? (BDM 7.10)

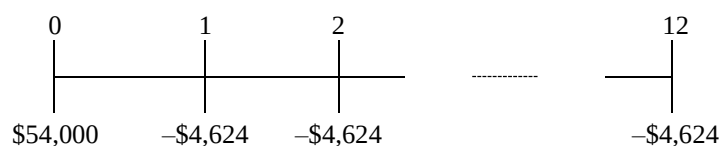
$$NPV = -\$500 + \frac{\$50,000}{1.16^{40}} = -\$367.98 < 0$$

$$\left(\frac{\$50,000}{\$500} \right)^{\frac{1}{40}} - 1 = 0.122 = 12.2\% < 16\%$$

Both rules agree – do not undertake the long-term investment opportunity.

- 5) Professor Wendy Smith has been offered the following deal: A law firm would like to retain her for an upfront payment of \$54,000. In return, for the next year the firm would have access to 8 hours of her time every month. Smith's rate is \$578 per hour and her opportunity cost of capital is 15% (per year). What does the IRR rule advise regarding this opportunity? What about the NPV rule? (BDM 7.13)

The timeline of this investment opportunity is:



$$NPV = 0 = \$54,000 - \left[\$4,624 \times \frac{1}{IRR} \left(1 - \frac{1}{(1 + IRR)^{12}} \right) \right]$$

Solving yields monthly IRR: 0.42%.

$$\text{Yearly IRR} = 1.0042^{12} - 1 = 0.05158 = 5.158\%.$$

Smith's cost of capital is 15%, so according to the IRR rule, she should turn down this opportunity.

Let's see what the NPV rule says. If you invest at an annual return of 15%, then after one month you will have $1.15^{\frac{1}{12}} - 1 = 1.171\%$, so the monthly cost of capital is 1.171%. Computing the NPV using this discount rate gives

$$NPV = \$54,000 - \left[\$4,624 \times \frac{1}{0.01171} \left(1 - \frac{1}{(1.01171)^{12}} \right) \right] = \$2,514.44$$

which is positive, so the correct decision is to accept the deal. Smith can also be relatively confident in this decision.

The reason that the IRR rule gives the opposite recommendation is that, in this case, positive cash flows occur first and negative cash flows later. When this happens, the naïve IRR rule should not be used, and instead the opposite recommendation should be followed (which is consistent with the NPV rule).

- 6) Your firm spends \$540,000 per year in regular maintenance of its equipment. Due to the economic downturn, the firm considers forgoing these maintenance expenses for the next three years. If it does so, it expects it will need to spend \$1.9 million in year 4 replacing failed equipment. (BDM 7.17)

- What is the IRR of the decision to forgo maintenance of the equipment?
- Does the IRR rule work for this decision?
- For what costs of capital is forgoing maintenance a good decision?

$$a. \quad 0 = \$540,000 \times \frac{1}{IRR} \left(1 - \frac{1}{(1+IRR)^3} \right) - \frac{\$1,900,000}{(1+IRR)^4}$$

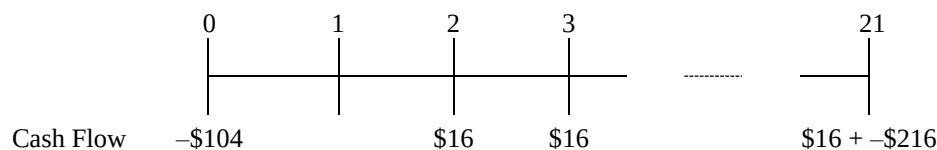
$$IRR = 8.19\%$$

- No, because positive cash flows precede negative ones.
- For all costs of capital $> IRR = 8.19\%$

- 7) You are considering constructing a new plant in a remote wilderness area to process the ore from a planned mining operation. You anticipate that the plant will take a year to build and cost \$104 million upfront. Once built, it will generate cash flows of \$16 million at the end of every year over the life of the plant. The plant will be useless 20 years after its completion once the mine runs out of ore. At that point you expect to pay \$216 million to shut the plant down and restore the area to its pristine state. Using a cost of capital of 12%, (BDM 7.20)

- What is the NPV of the project?
- Is using the IRR rule reliable for this project? Explain.

Timeline:



$$a. \quad NPV = -\$104 + \frac{\left[\$16 \times \frac{1}{0.12} \left(1 - \frac{1}{(1.12)^{20}} \right) \right]}{1.12} - \frac{\$216}{(1.12)^{21}}$$

$$NPV = -\$104 + \$106.71 - \$19.99 = -\$17.28 \text{ million}$$

- No, the IRR rule is not reliable because the project has a negative cash flow that comes after the positive ones.

The Payback Rule

- 8) You are a real estate agent thinking of placing a sign advertising your services at a local bus stop. The sign will cost \$7,000 and will be posted for one year. You expect that it will generate additional revenue of \$1,120 per month. What is the payback period? (BDM 7.21)

$$\text{Payback period} = \frac{\$7,000}{\$1,120} = 6.25 \text{ months}$$

Choosing between Projects

No question.

Project Selection with Resource Constraints

- 9) You own a car dealership and are trying to decide how to configure the showroom floor. The floor has 2,000 square feet of usable space. You have hired an analyst and asked her to estimate the NPV of putting a particular model on the floor and how much space each model requires:

Model	NPV	Space Requirement (sq. ft.)
MB345	\$4000	200
MC237	4900	250
MY456	3000	240
MG231	4000	150
MT347	6000	450
MF302	2000	200
MG201	2500	150

In addition, the showroom also requires office space. The analyst has estimated that office space generates an NPV of \$14 per square foot. What models should be displayed on the floor and how many square feet should be devoted to office space?

(BDM 7.32)

Model	NPV	Space Requirement (sq.	NPV/sq.
MB345	\$4,000	200	\$20
MC237	\$4,900	250	\$20
MY456	\$3,000	240	\$13
MG231	\$4,000	150	\$27
MT347	\$6,000	450	\$13
MF302	\$2,000	200	\$10
MG201	\$2,500	150	\$17

Take the MG231, MB345, MC237, and MG201 (750 sq. ft)
Use remaining 1,250 sq.ft for office space.

- 10) Orchid Biotech Company is evaluating several development projects for experimental drugs. Although the cash flows are difficult to forecast, the company has come up with the following estimates of the initial capital requirements and NPVs for the projects. Given a wide variety of staffing needs, the company has also estimated the number of research scientists required for each development project (all cost values are given in millions of dollars). (BDM 7.34)

Project Number	Initial Capital	Number of Research Scientists	NPV
I	\$10	2	\$10.1
II	15	3	19.0
III	15	4	22.0
IV	20	3	25.0
V	30	12	60.2

- Suppose that Orchid has a total capital budget of \$60 million. How should it prioritize these projects?
- Suppose in addition that Orchid currently has only 12 research scientists and does not anticipate being able to hire any more in the near future. How should Orchid prioritize these projects?

<u>Project</u>	<u>PI</u>	<u>NPV/Headcount</u>
I	1.01	5.1
II	1.27	6.3
III	1.47	5.5
IV	1.25	8.3
V	2.01	5.0

- The PI rule selects projects V, III, II. These are also the optimal projects to undertake (as the budget is used up fully taking the projects in order).
- The PI rule using the headcount constraint alone selects IV, II, III, I, and V, because the project with the next highest PI (that is NPV/Headcount), V, cannot be undertaken without violating the resource constraint. These projects are also feasible to do under the current capital budget because they happen to require exactly \$60 million in capital. The only other feasible possibility is to take only project V, which generates a lower NPV, so this choice of projects is optimal.

Exercises Chapter 3: Bonds**Additional Exercises****Bond Cash Flows, Prices, and Yields**

- 1) Assume that a bond will make payments as shown on the following timeline: (BDM 6.2)



- a. What is the maturity of the bond (in years)?
 - b. What is the coupon rate (in percent)?
 - c. What is the face value?
- a. The maturity is 20 years.
 - b. $(\$30/\$1,000) = 3\%$, so the coupon rate is 3%.
 - c. The face value is \$1,000.

Dynamic Behavior of Bond Prices

- 2) Consider the following bonds with a face value of \$100: (BDM 6.13)

Bond	Coupon Rate (annual payments)	Maturity (years)
A	0%	16
B	0%	8
C	4%	16
D	7%	8

- a. What is the percentage change in the price of each bond if its yield to maturity falls from 6% to 5%?
 - b. Which of the bonds A–D is most sensitive to a 1% drop in interest rates from 6% to 5% and why? Which bond is least sensitive? Provide an intuitive explanation for your answer.
- a. We can compute the price of each bond at each YTM.

$$P(\text{bond A}, 6\% \text{ YTM}) = \frac{\$100}{1.06^{16}} = \$39.36$$

$$P(\text{bond B}, 6\% \text{ YTM}) = \frac{\$100}{1.06^8} = \$62.74$$

$$P(\text{bond C}, 6\% \text{ YTM}) = \$4 \times \frac{1}{0.06} \left(1 - \frac{1}{1.06^{16}}\right) + \frac{\$100}{1.06^{16}} = \$79.79$$

$$P(\text{bond D}, 6\% \text{ YTM}) = \$7 \times \frac{1}{0.06} \left(1 - \frac{1}{1.06^8}\right) + \frac{\$100}{1.06^8} = \$106.21$$

We calculate the price of each bond in the same fashion when the YTM = 5%. Once we compute the price of each bond for each YTM, we can compute the % price change as

$$\text{Percent change} = \frac{(\text{Price at 5\% YTM}) - (\text{Price at 6\% YTM})}{(\text{Price at 6\% YTM})}$$

The results are shown in the table below.

Par value	1.000,00				
Yield to maturity	6,00%				
Bond	Coupon Rate	Maturity	Price	Price at 5,00%	Percentage Change
A	0,00%	16	393,65	458,11	16,38%
B	0,00%	8	627,41	676,84	7,88%
C	4,00%	16	797,88	891,62	11,75%
D	7,00%	8	1.062,10	1.129,26	6,32%

- b. Bond A is most sensitive, because it has the longest maturity and no coupons, so its future cash flows have the highest discount factors. Bond D is the least sensitive. Intuitively, higher coupon rates and a shorter maturity mean that relatively more of the bond's cash flows happen early and thus cannot be as greatly affected by changes in interest rates as bonds with low coupon rates and longer maturities.

Investing in Bonds

- 3) NETWORK AG issues a fixed coupon bond with the following terms: (InvFin exam 17/18_1)

- Face value: €100
- Coupon rate: 6%
- Time to maturity: 5 years

- The current market return of similar bonds is 4 % per year. What would you be willing to pay for the bond?
- You bought the bond immediately after issuance at the fair market price determined in a) and sell it again after three years. In the meantime, the market interest rate has risen to 5%. What selling price do you achieve when you sell? What is the IRR that you earned during the three years?
- What IRR would you have earned if you had held the bond until maturity after 5 years?

a. The market value is assumed to be the present value, which is €108.90.

b. The market value for the remaining two years is assumed to be the present value, which is €101.86.

Thus, the investor paid €108.90 for a cash flow stream of €6 at the end of year 1, €6 at the end of year 2, and €6 + €101.86 at the end of year 3. The IRR of this stream is 3.41%.

- c. The IRR would be at 4% per year, i.e., the market interest rate as of your initial investment. No calculations needed.

Corporate Bonds

- 4) Andrew Industries is contemplating issuing a 30-year bond with a coupon rate of 9.69% (annual coupon payments) and a face value of \$1,000. Andrew believes it can get a rating of A from Standard and Poor's. However, due to recent financial difficulties at the company, Standard and Poor's is warning that it may downgrade Andrew Industries bonds to BBB. Yields on A-rated, long-term bonds are currently 9.19%, and yields on BBB-rated bonds are 9.59%. (BDM 6.30)

- a. What is the price of the bond if Andrew maintains the A rating for the bond issue?
- b. What will the price of the bond be if it is downgraded?

- a. When originally issued, the price of the bond was

$$P = \$96.9 \times \frac{1}{0.0919} \left(1 - \frac{1}{1.0919^{30}} \right) + \frac{\$1,000}{1.0919^{30}} = \$978.98 + \$71.53 = \$1,050.51$$

- b. If the bond is downgraded, its price will fall to

$$P = \$96.9 \times \frac{1}{0.0959} \left(1 - \frac{1}{1.0959^{30}} \right) + \frac{\$1,000}{1.0959^{30}} = \$945.66 + \$64.10 = \$1,009.76$$

Sovereign Bonds

No question.

Exercises Chapter 4: Stocks**Additional Exercises****Dividend-Discount Model (DDM)**

- 1) Krell Industries has a share price of \$21.44 today. If Krell is expected to pay a dividend of \$0.78 this year, and its stock price is expected to grow to \$24.63 at the end of the year, what is Krell's dividend yield and equity cost of capital? (BDM 9.4)

$$\text{Dividend yield} = \frac{\text{Div}_1}{P_0}$$

$$\text{Dividend yield} = \frac{\$0.78}{\$21.44} = 0.0364 = 3.64\%$$

$$\text{Capital gain} = \frac{P_1 - P_0}{P_0}$$

$$\text{Capital gain} = \frac{\$24.63 - \$21.44}{\$21.44} = 0.1488 = 14.88\%$$

$$\text{Equity cost of capital} = 3.64\% + 14.88\% = 18.52\%$$

Applying the DDM

- 2) Dorpac Corporation has a dividend yield of 1.7%. Dorpac's equity cost of capital is 7.4%, and its dividends are expected to grow at a constant rate. (BDM 9.7)

a. What is the expected growth rate of Dorpac's dividends?

b. What is the expected growth rate of Dorpac's share price?

a. $r_E = \frac{\text{Div}_1}{P_0} + g$

$$7.4\% = 1.7\% + g$$

$$g = 7.4\% - 1.7\% = 5.7\%$$

b. With constant dividend growth, share price is also expected to grow at rate $g = 5.7\%$ (or we can solve this from Eq 9.2).

$$r_E = \frac{\text{Div}_1 + P_1}{P_0} - 1 = \underbrace{\frac{\text{Div}_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$$

- 3) Cooperton Mining just announced it will cut its dividend from \$4.27 to \$2.67 per share and use the extra funds to expand. Prior to the announcement, Cooperton's dividends were expected to grow at a 2.9% rate, and its share price was \$49.06. With the new expansion, Cooperton's dividends are expected to grow at a 4.8% rate. What share price would you expect after the announcement? (Assume Cooperton's risk is unchanged by the new expansion.) Is the expansion a positive NPV investment? (BDM 9.11)

$$r_E = \frac{Div_1}{P_0} + g$$

$$r_E = \frac{\$4.27}{\$49.06} + 0.029 = 0.116 = 11.6\%$$

$$P_t = \frac{Div_t(1+g)}{r-g}$$

$$P_t = \frac{\$2.67}{0.116-0.048} = \$39.26$$

In this case, cutting the dividend to expand is not a positive NPV investment.

- 4) Colgate-Palmolive Company has just paid an annual dividend of \$1.35. Analysts are predicting dividends to grow by \$0.12 per year over the next five years. After then, Colgate's earnings are expected to grow 6.7% per year, and its dividend payout rate will remain constant. If Colgate's equity cost of capital is 8.6% per year, what price does the dividend-discount model predict Colgate stock should sell for today? (BDM 9.13)

$$P_0 = \frac{Div_1}{1+r_E} + \frac{Div_2}{(1+r_E)^2} + \dots + \frac{Div_N}{(1+r_E)^N} + \frac{1}{(1+r_E)^N} \left(\frac{Div_{N+1}}{r_E - g} \right)$$

$$P_0 = \frac{\$1.35+0.12}{1.086} + \frac{\$1.47+0.12}{(1.086)^2} + \frac{\$1.59+0.12}{(1.086)^3} + \frac{\$1.71+0.12}{(1.086)^4} + \frac{\$1.83+0.12}{(1.086)^5} + \frac{1}{(1.086)^5} \left(\frac{\$1.95(1.067)}{0.086-0.067} \right)$$

$$P_0 = \$1.35 + \$1.35 + \$1.34 + \$1.32 + \$1.29 + \$72.49 = \$79.14$$

- 5) Procter and Gamble (PG) paid an annual dividend of \$2.87 in 2018. You expect PG to increase its dividends by 8% per year for the next five years (through 2023), and thereafter by 3% per year. If the appropriate equity cost of capital for Procter and Gamble is 8% per year, use the dividend-discount model to estimate its value per share at the end of 2018. (BDM 9.12)

Since dividends are growing at the cost of capital over the next 5 years, the present value of the next 5 years' dividends are:

$$PV_{1-5} = \frac{2.87 \cdot 1.08}{1.08} + \frac{2.87 \cdot 1.08^2}{1.08^2} + \dots + \frac{2.87 \cdot 1.08^5}{1.08^5} = 5 \cdot 2.87 = \$14.35.$$

Value on date 5 of the rest of the dividend payments:

$$PV_5 = \frac{2.87(1.08)^5 \cdot 1.03}{0.08 - 0.03} = 86.87$$

Discounting this value to the present gives

$$PV_0 = \frac{86.87}{(1.08)^5} = \$59.12$$

So the value of P&G is: $P = PV_{1-5} + PV_0 = 14.35 + 59.12 = \73.47 .

Market Efficiency

- 6) In early 2018, Coca-Cola Company (KO) had a share price of \$42.42, and had paid a dividend of \$1.01 for the prior year. Suppose you expect Coca-Cola to raise this dividend by approximately 6.6% per year in perpetuity. (BDM 9.31)
- If Coca-Cola's equity cost of capital is 8.3%, what share price would you expect based on your estimate of the dividend growth rate?
 - Given Coca-Cola's share price, what would you conclude about your assessment of Coca-Cola's future dividend growth?
- $\frac{\$1.01(1.066)}{(0.083 - 0.066)} = \63.33
 - Based on the market price, our growth forecast is probably too high. Growth rate consistent with market price is $g = r_E - \text{dividend yield} = 8.3\% - [(1.01 \times 1.066) / 42.42] = 5.76\%$, which is more reasonable.
- 7) You read in the paper that Summit Systems from Basic Exercises Problem 5 has revised its growth prospects and now expects its dividends to grow at 3% per year forever. (BDM 9.30)
- What is the new value of a share of Summit Systems stock based on this information?
 - If you tried to sell your Summit Systems stock after reading this news, what price would you be likely to get and why?
- $P = 1.50 / (11\% - 3\%) = \18.75 .
 - Given that markets are efficient, the new growth rate of dividends will already be incorporated into the stock price, and you would receive \$18.75 per share. Once the information about the revised growth rate for Summit Systems reaches the capital market, it will be quickly and efficiently reflected in the stock price.
- 8) Which group of investors is capable of earning consistent, superior profits if financial markets are strong-form efficient? Technical analysts, fundamental analysts or insider traders? (BMM 120)
- No one will be capable of sustained, superior profits.
- 9) How does competition among investors lead to efficient markets? (BMM 140)

The search for information and insightful analysis makes investor assessments of stock values as reliable as possible. Since the rewards accrue to the investors who uncover relevant information before

it is reflected in stock prices, competition among these investors means that there is always an active search for mispriced stocks.

Competition between investors will tend to produce an efficient market—that is, a market in which prices rapidly reflect new information, and investors have difficulty making consistently superior returns. Of course, we all hope to beat the market, but if the market is efficient, all we can rationally expect is a return that is sufficient on average to compensate for the time value of money and for the risks we bear.

- 10) Your parents who are not very familiar with business topics turn to you and ask whether they should invest some money in the DAX because the chart looks very convincing. As a business student in the second semester, they expect that you can make a profound statement how the DAX will develop within the next twelve months. What do you answer them? (Extra Question)



DAX 1.1.2019-20.2.2020; taken from www.comdirect.de

Many different directions for discussion available, e.g.: Skepticism whether future share prices could be forecasted; could a single person beat the aggregated “wisdom” of all market participants; exploitability of trends; does a long period of rising prices necessarily mean that shares are overvalued now (what goes up must come down); ...

Exercises Chapter 5: Capital Asset Pricing Model**Additional Exercises****Diversification**

- 1) Calculate the expected return, variance, and standard deviations for investments in either stock A or stock B, or an equally weighted portfolio of both. (BMM 11.110)

Scenario	Probability	Return on A	Return on B
Recession	25%	-4%	9%
Normal	40%	8%	4%
Boom	35%	20%	-4%

Stock A:

Expected return = $(.25 \times -4\%) + (.40 \times 8\%) + (.35 \times 20\%) = 9.2\%$

Variance = $.25(-4\% - 9.2\%)^2 + .4(8\% - 9.2\%)^2 + .35(20\% - 9.2\%)^2 = 43.56 + .58 + 40.82 = 84.96$

Standard deviation = $\text{SQRT}(84.96) = 9.22\%$

Stock B:

Expected return = $(.25 \times 9\%) + (.40 \times 4\%) + (.35 \times -4\%) = 2.45\%$

Variance = $.25(9\% - 2.45\%)^2 + .4(4\% - 2.45\%)^2 + .35(-4\% - 2.45\%)^2 = 10.73 + .96 + 14.56 = 26.25$

Standard deviation = $\text{SQRT}(26.25) = 5.12\%$

Portfolio:

Expected return = $(.5 \times 9.2\%) + (.5 \times 2.45\%) = 5.83\%$

Variance = $.25(((-4\% + 9\%) \times .5) - 5.83\%)^2 + .4(((8\% + 4\%) \times .5) - 5.83\%)^2 + .35((20\% - 4\%) \times .5) - 5.83\%)^2 = 2.77 + .01 + 1.65 = 4.43$

Standard deviation = $\text{SQRT}(4.43) = 2.1\%$

Note that the standard deviation of the portfolio is considerably less than that of either Stock A or Stock B.

- 2) Discuss the statement, "Only market risk matters to a diversified investor." (BMM 11.112)

It is relatively easy for an investor to diversify a portfolio sufficiently to the point where nearly all of the remaining risk is market risk. This can be achieved by combining somewhere between 15 and 20 different securities. At that point, then, the investor does not care about a strike at General Motors because that element of unique risk has been offset by a boom in the semiconductor industry. Any investor who chooses not to diversify and thus retain the unique risk should be prepared to suffer the consequences; basically, risk level will increase without a corresponding increase in expected yield. These statements should show that diversified investors are concerned only with the market risk in their portfolios, rather than any elements of unique risk.

Total Risk and Market Risk

- 3) Explain why an insurance company is willing to sell life insurance to individuals, but will be more reluctant to issue policies insuring against earthquake damage to residents living along fault lines. Why don't insurance companies simply charge the residents a premium that reflects the actuarial probability of damage from earthquakes? (BMM 12.TB122)

The risks of deaths of individual policyholders are largely independent, and are therefore diversifiable. The insurance company is thus satisfied to charge a premium that reflects the actuarial probabilities of death, without an additional risk premium. In contrast, earthquake damage is not independent across policyholders. If someone's house is damaged by an earthquake, there is a greater chance that the neighbor's house will be damaged too. Because earthquake risk is not as diversifiable, the insurance company may not be satisfied to charge a premium that reflects only the expected value of payouts.

- 4) Why is beta thought to be a more relevant measure of risk than standard deviation for a diversified investor? (BMM 12.TB115)

Standard deviation measures both a stock's market risk and unique risk. However, a diversified investor is no longer concerned with unique risk, or at least not concerned over the small portion that remains after portfolio diversification. Beta measures only the market risk of the stock, that type of risk that cannot be diversified away. The stock's returns may be more or less volatile than the market portfolio and beta is an indication of that sensitivity.

Portfolio Beta

- 5) An investor divides her portfolio into thirds, with one part in Treasury bills, one part in a market index, and one part in a diversified portfolio with beta of 1.50. What is the beta of the investor's overall portfolio? (BMM 12.TB88)

$$\beta_{Portfolio} = .33 \times 0 + .33 \times 1 + .33 \times 1.5 = 0.833$$

SML and CAPM

- 6) What return would be expected by an investor whose portfolio was 25% market portfolio and 75% Treasury bills if the risk-free rate was 7% and the market risk premium was 8%? (BMM 12.TB67)

Expected return on market = r_f + market risk premium = 15.00%

Expected return on portfolio = $(.75 * 7\%) + (.25 * 15\%) = 5.25\% + 3.75\% = 9.00\%$

- 7) The manager of StarPerformer Mutual Fund expects the fund to earn a rate of return of 12% this year. The beta of the fund's portfolio is .8. If the rate of return available on risk-free assets is 5% and you expect the rate of return on the market portfolio to be 15%, should you invest in StarPerformer? Can you create a portfolio with the same risk as StarPerformer Mutual Fund, but with a higher expected rate of return?

(BMM 12.TB124)

The CAPM implies that the expected rate of return that investors will demand of the portfolio is:

$$r = r_f + \beta * (r_m - r_f) = 5\% + 0.8 * (15\% - 5\%) = 13\%$$

If the portfolio is expected to provide only a 12% rate of return, it is an unattractive investment. The portfolio does not provide an expected return that is sufficiently high relative to its risk.

A portfolio that is invested 80% in a stock index mutual fund (with a beta of 1.0) and 20% in Treasury bills or a money market mutual fund (with a beta of zero) would have the same beta as StarPerformer Mutual Fund: $\beta = (0.80 * 1.0) + (0.20 * 0) = 0.80$

However, the portfolio will provide an expected return of: $(0.80 * 15\%) + (0.20 * 5\%) = 13\%$

This is better than the expected return for StarPerformer Mutual Fund.

Exercises Chapter 6: Capital Market Based Cost of Capital**Additional Exercises****Applying the CAPM in Capital Budgeting**

- 1) Assume the Treasury bill rate is 4% and the market risk premium is 7%. What are the project costs of capital for new ventures with betas of 0.75 and 1.75? (BMM 11.PP12b)

Beta	Cost of Capital (from CAPM)
0.75	$4\% + (0.75 * 7\%) = 9.25\%$
1.75	$4\% + (1.75 * 7\%) = 16.25\%$

- 2) Why is it important to make the distinction between company opportunity cost of capital and project opportunity cost of capital when evaluating projects? (BMM 11.TB121)

The company opportunity cost of capital represents an average risk of the individual projects currently invested in by the company. It is thus possible that no one project exemplifies this "average" company risk. Unless the proposed project is coincidentally of the same level of risk as the company opportunity cost of capital, it would be inappropriate to use the company discount rate. Rather, the merit (e.g., NPV) of the project should be judged using the level of risk that pertains to the project itself. As an example, it would be incorrect to use a low company cost of capital to determine the NPV of a project that has a high degree of risk. This offers a mistakenly high estimate of NPV that is not likely to materialize.

Weighted-Average Cost of Capital (WACC)

- 3) What is the WACC for a firm using 55% equity with a required return of 15%, 45% debt with a required return of 8%, and a tax rate of 35%? (BMM 13.40)

$$\text{WACC} = [.45 * (1 - .35).08] + (.55 * .15) = 2.34\% + 8.25\% = 10.59\%$$

- 4) For a company that pays no corporate taxes, its WACC will be equal to: (BMM 13.58)
- the expected return on its assets.
 - the expected return on its debt.
 - the total value of its assets.
 - the expected return on its equity.

a.

- 5) What is the WACC for a firm with 60% debt and 40% equity if the respective costs for these components are 9.23% before tax and 18% before tax? The firm's tax rate is 35%. (BMM 13.78)

$$\text{WACC} = [.6 * (1 - 35\%) * .0923] + (.4 * .18) = .036 + .072 = 10.8\%$$

- 6) Plasti-tech Inc. is financed 60% with equity and 40% with debt. Currently, its debt has a before-tax interest rate of 12%. Plasti-tech's common stock trades at \$15.00 per share and its most recent dividend was \$1.00. Future dividends are expected to grow by 4%. If the tax rate is 34%, what is Plasti-tech's WACC? (BMM 13.90)

$$r_e = (1 * (1 + 0.04)) / \$15 + 0.04 = 10.93\%$$

$$\text{WACC} = 0.4 * [0.12 * (1 - 0.34)] + 0.6 * (0.1093) = .0317 + .0656 = 9.73\%$$

- 7) How should the weighted-average costs of capital be applied to projects that are not average? (BMM 13.100)

The weighted-average cost of capital can still be used as a benchmark. The benchmark is adjusted up for unusually risky projects and down for unusually safe ones.

- 8) Why is it important to use market values rather than book values when determining the weighted-average cost of capital? (BMM 13.101)

Book values represent the amount and terms of funding for a firm or a project as it was originally placed into existence. Because firms and projects are often long-lived, circumstances can change current values substantially from their original book values. These current, or market, values represent the consensus of the investment community as to the value of the firm or project at this point in time. Recalling that past values may represent sunk costs, it is imperative that values be determined with an eye toward their future benefits.